PUSAN NATIONAL UNIVERSITY

Dept. of Aerospace Engineering

Computational Structural Analysis

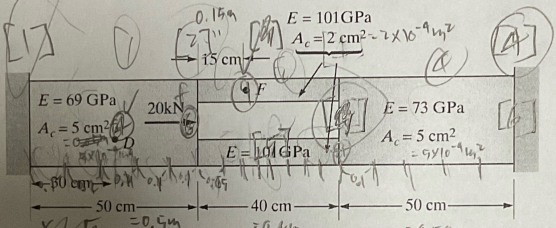
Mid-exam. Oct. 24th, 2023

NOTICE: Write down the ANSYS commands on your answer sheet.

Problem 1

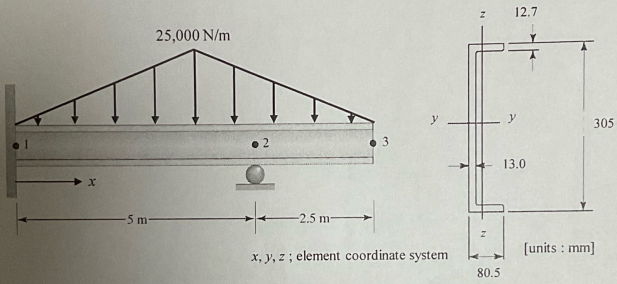
(a) Determine the deflections of point *D* and point *F* in the below system.

(b) Compute the axial force and stress in each member.



Problem 2

Consider a beam in the figure below, which has a cross-section in the shape of a C-channel (C310 × 45), as shown in the right figure. The beam is made of steel with a Young's Modulus of 200 GPa and is subjected to a gradually varying distributed load, reaching a maximum value of 25,000 N/m at the center of the beam.



(a) Obtain the reaction forces at the position #1

(b) Check for the validity of FEA results using the static equilibrium conditions.

(c) Determine the vertical displacement and rotation at the position #3 and the rotation at the position #2.

(d) When a linear elastic spring instead of the fixed B.C is attached at the position #2, as shown in below figure, what is the minimum value of the spring constant, k, required to ensure that the nodal displacement at position #2 is 1mm or less? Use element type of LINK180 to model the steel spring.

